

X-ray computed tomography-based characterisation of graphene nanoplatelets re-agglomeration in hardened cement composites

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ABSTRACT

Graphene Nanoplatelets (GNPs) have been found to be an effective additive for enhancing the mechanical strength of cementitious materials, attributed to the role they play as nucleation sites. However, there is not a direct correlation on the enhancing effect of graphene with increasing graphene dosage, resulting in different optimal dosages for different binders and mix designs. This study aims to develop a method to clarify the governing factor that determines the optimal graphene dosage in mortar mixes. X-ray computed tomography (XCT) was used to identify the size, surface area and distribution/re-agglomeration of graphene nanoplatelets in the prepared mixes. Compressive strength tests and TGA analysis were carried out to evaluate correlations among the total surface area of graphene nanoplatelets, mechanical performance, and cement hydration. Out of the three graphene dosages (0.035, 0.07, and 0.1 wt% by mass of cement) used in this study, samples with 0.07 wt% graphene revealed the highest total surface area of graphene nanoplatelets, hydration degree, and compressive strength. Results revealed that the enhancing effect of graphene addition in cement-based mixes was associated with the total surface area of the graphene nanoplatelets. A larger surface area contributed to stronger mechanical reinforcement by providing an increased number of nucleation sites in the pore solution.

1. Introduction

Concrete is the most commonly used construction material in the world. Several studies confirmed the beneficial effects of graphene on the properties of cement-based mixes. Inclusion of small amounts of graphene (i.e. as low as 0.04 wt% of cement mass) into the cement matrix can significantly increase the compressive and flexural strengths of pastes by up to 30 % and 84 %, respectively [1]. Other potential benefits include increased thermal and electrical conductivity, chloride diffusion resistance, and self-sensing [2–6]. Generally, most of these benefits are realized through the nucleation seeding effect achieved via the addition of graphene nanoplatelets (GNPs) to cement mixes, resulting in an acceleration of the cement hydration process [1,2,5,7,8]. Furthermore, GNPs can also fill voids, arrest micro-cracks and generate interlocking among hydration products [1,5,9]. As a part of this mechanism, the accelerated hydration products can form on the surface of GNPs; thus, the total surface area (TSA) of GNPs in hardened cement paste can play a critical role in hydration enhancement [10–12]. However, a quantitative investigation of the correlation between the TSA of GNPs and strength development has not been reported yet. In this

respect, TSA could be used as an indicator to determine the optimum dosage of GNPs for strength enhancement, promoting efficient material usage.

One of the main challenges regarding the use of GNPs is their tendency to re-agglomerate in liquid and viscous materials, such as cement pastes. GNPs re-agglomeration reduces the surface area of graphene as nucleation sites, limiting the efficiency of graphene in enhancing the properties of cementitious materials as the degree of re-agglomeration increases. As illustrated in Fig. 1, re-agglomeration of GNPs (Fig. 1(b)) can lead to a drastic decrease in the surface area of GNPs within the cement matrix, compared to well-dispersed status (Fig. 1(a)). Therefore, to maximize the potential of GNPs in cementitious materials, the degree of re-agglomeration must be minimized [13–15]. In order to disperse GNPs homogeneously and minimise re-agglomeration, several strategies have been commonly adopted, including ultrasonication treatment, mechanical stirring, introducing surfactant, and functionalization [14–18].

However, regardless of the mixing method, GNPs re-agglomeration cannot be eliminated [19–22]. The degree of re-agglomeration increases with increasing graphene dosage, resulting in an optimal level of graphene dosage, beyond which further increases decrease the

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Glossary:

GNPs	Graphene nanoplatelets
TSA	Total surface area
TSA ₀	Total surface area per unit volume of cement composite
XCT	X-ray computed tomography
TGA	Thermogravimetric analysis
SEM	Scanning electron microscope
F-value	FeretShape value

enhancement of mechanical properties, as summarized in Table 1. However, existing investigations (as listed in Table 1) on the mechanisms of GNPs enhancing the mechanical properties of cementitious materials are generally qualitative, often based on mechanical test results or scanning electron microscope (SEM) observations, which can only show superficially exposed GNPs. Furthermore, within the limited field SEM offers, the embedded GNPs would not be clearly observable, making it difficult to use SEM images (e.g., Fig. 2) to comprehensively quantify the surface areas of GNPs within an entire sample [22,23].

Alternatively, a more detailed quantitative characterisation of GNPs in hardened cementitious materials, including their actual morphology, size distribution, spatial dispersion and surface areas can be achieved

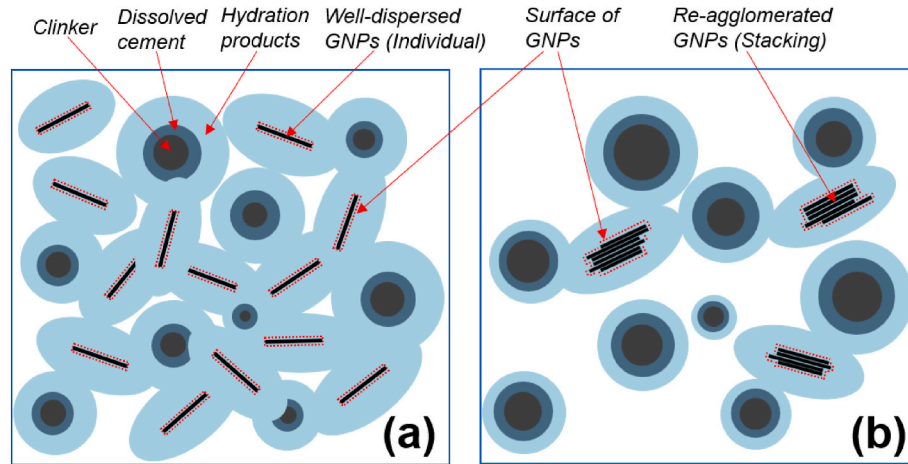


Fig. 1. Illustration of re-agglomeration of GNPs: (a) well-dispersed status of GNPs offering seeding effect; (b) re-agglomerated status of GNPs with less seeding effect.

Table 1

Summary of experimental results on the mechanical properties of cement and concrete with different graphene dosages.

References	Type of mixture	Mechanical strength	Optimal dosage (wt. %)	Maximum enhancement (%)
[1]	Cement	Compressive	0.02	38.7
	Paste	Flexural	0.02	38.0
[5]	Mortar	Compressive	0.05	8.3
		Flexural	0.05	15.6
[14]	Cement	Compressive	0.01	13.5
	paste	Tensile	0.01	7.6
[20]	Concrete	Compressive	0.05	22.4
[24]	Mortar	Compressive	0.3	10.3
[25]	Concrete	Compressive	0.025	16.8
[26]	Mortar	Compressive	0.05	32.4
[27], ^a	Cement	Flexural	0.03	36.6
	Paste			
[28]	Cement	Compressive	0.06	11.2
	Paste	Flexural	0.06	28.2
[29]	Mortar	Compressive	0.08	43.3
		Flexural	0.08	22.2
[30]	Mortar	Compressive	0.07	34.3
		Tensile	0.07	26.9
[31]	Cement	Compressive	0.05	3.7
	Paste	Flexural	0.05	25.2
[32]	Cement	Compressive	0.06	11.2
	Paste	Flexural	0.06	20.0
[33]	Cement	Compressive	0.06	6
	Paste	Flexural	0.06	6

^a All the mechanical properties were investigated at a curing age of 28 days in all studies presented in this table, except for ref [28], where results were reported at 7 days. The fillers were all GNPs.

through more comprehensive characterisation techniques, such as X-ray computed tomography (XCT). Although a few studies have used XCT imaging to qualitatively investigate the effects of graphene on cementitious materials [21,23,34–37], there is currently a gap in the literature regarding the quantitative characterisation of GNPs re-agglomeration and its correlation with the enhancement of mechanical properties of cementitious mixes.

Amongst other characteristics, the total surface area of graphene plays a key role in the enhancement of the hydration of cement via the provision of additional nucleation sites [14,26,38,39]. Therefore, it is important to identify the graphene dosage at which the total graphene surface area reaches the maximum value, which could be correlated with the optimal graphene dosage in a cementitious mix. Furthermore, once the link between total graphene surface area and improvement in mechanical performance is confirmed, a predictive numerical simulation model can be developed to calculate GNPs re-agglomeration and reveal the optimal graphene dosage in cementitious materials [40].

Based on these current gaps in the literature, the objectives of this study are to: (i) provide quantitative evidence to confirm the positive correlation between the total surface area (TSA) of GNPs and the enhancement in mechanical performance of cementitious mixes; and (ii) provide quantitative information on sizes and distributions of GNPs to inform future numerical modelling of graphene re-agglomeration in cementitious materials. To achieve these research objectives, a comprehensive XCT image analysis was carried out, revealing the total surface area per unit volume of cement composite (TSA₀) and size distribution of GNPs. Changes in the total surface area at different graphene dosages were then compared to the enhancements of hydration degree and compressive strength in the prepared mortar samples. Conclusions were drawn regarding the relationship among GNPs dosage, level of re-agglomeration in the cement matrix, total surface area considering re-agglomeration and the influence of these on hydration degree and

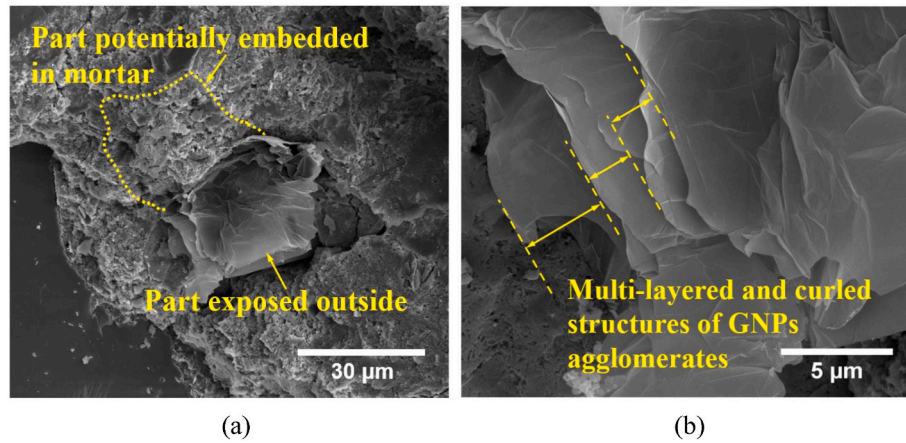


Fig. 2. SEM images showing: (a) GNPs embedded in mortar, and (b) re-agglomeration of GNPs in hardened mortar.

Table 2
Properties of graphene used in this study.

Type of graphene	Purity	Lateral size	Layer number	Specific surface area	Tapped Density
Dry graphene nanoplatelets	99 %	20 μm	87	0.65 m^2/cm^3	0.395 g/cm^3

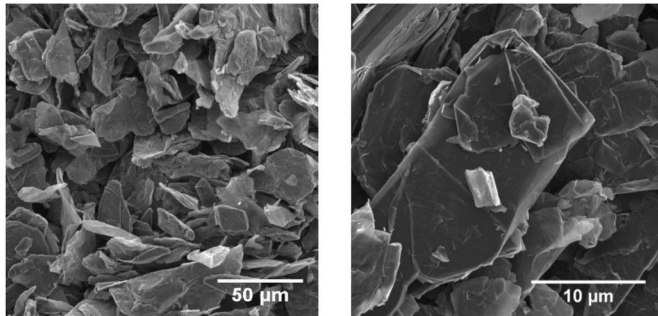


Fig. 3. Typical SEM images of the as-received dry GNPs.

mechanical properties. The obtained results enabled the identification/prediction of the optimal dosage of GNPs in the cement matrix.

2. Materials and methodology

2.1. Materials

Commercial GNPs with a purity of >99 %, provided by First Graphene (UK) Ltd were used. The average diameter of the as-received graphene nanoparticles was 20 μm . Table 2 lists the key information of the graphene. Fig. 3 shows typical SEM images of the dry GNPs, revealing the wrinkled surfaces, as well as folded and multi-layered forms.

Standard sand as specified in ASTM C778 [41] and Portland Cement (PC) 42.5N were used. Table 3 lists the chemical compositions of PC.

Table 3
Chemical composition of cement (%) used in this study.

Type of cement	CaO	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	SO ₃	K ₂ O	In ₂ O ₃	TiO ₂
PC 42.5N	71.133	13.612	3.547	3.328	5.037	0.519	1.587	0.303

2.2. Preparation of graphene suspension and cement mixture

The same process of preparing graphene suspension as reported by Ref. [42] was used in the present study. While ultrasonic treatment, with the aid of surfactant, effectively dispersed graphene in water, agglomeration could not be entirely eliminated, particularly at higher dosages [1,19–22]. Before mixing with cement, GNPs were dispersed in tap water. Dry GNPs were first mixed with tap water and surfactant (poly-carboxylic ether polymer-based superplasticizer ADVA 650 (SP)), and then hand stirred for 2 min. Afterwards, the mixture was subjected to ultrasonic treatment (Cole-Palmer Ultrasonic Processor Type CP505, 30 % amplitude of the maximum power 500W) for 30 min. The ultrasonicated GNPs suspension was then mixed with cement powder, sand and tap water to produce fresh cement mortar, followed by 2 min of low-speed stirring and 3 min of high-speed stirring.

In this research, three different graphene dosages (i.e., G1 = 0.035 wt%, G2 = 0.07 wt% and G3 = 0.1 wt% by mass of cement) were investigated, in addition to the reference mixture without graphene (G0). The water to cement ratio was set at 0.4 for all specimens. The fresh mortar mixture was cast into cubic moulds of dimensions of 50 × 50 × 50 mm^3 and consolidated by a vibrating table for 30 s. All the specimens were covered with plastic sheets and demoulded after 24 h. Afterwards, the specimens were stored in saturated lime water to provide a stabilized pH environment and mitigated shrinkage of matrix at ambient temperature for 28 days for curing before any test. Three samples were prepared for compressive strength test for each mix. Table 4 lists the mix proportions, which were based on previous research studies by others [30,42].

Table 4
Mixing proportions of specimens with different graphene dosages.

Sample No	GNPs dosage (wt.%)	Cement (kg/ m^3)	Sand (kg/ m^3)	Water (kg/ m^3)	SP (kg/ m^3)
G0 (Ref)	0	1000	1400	400	10
G1	0.035	1000	1400	400	10
G2	0.07	1000	1400	400	10
G3	0.1	1000	1400	400	10

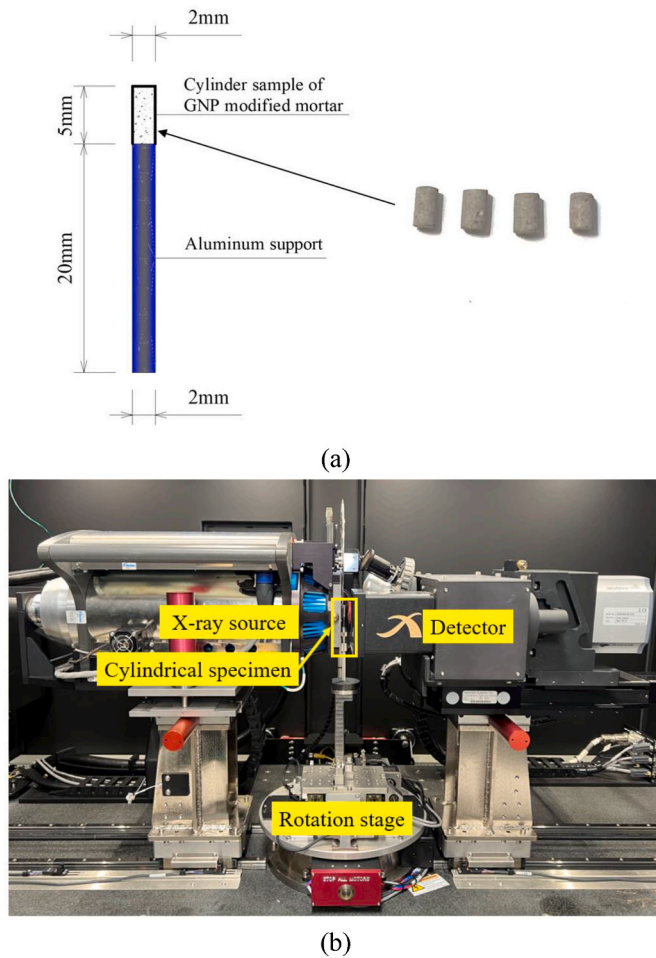


Fig. 4. XCT scanning setup, showing (a) dimensions of cement composite cylindrical specimens; and (b) internal configuration of XCT instrument.

2.3. X-ray computed tomography scanning

XCT scanning for characterising the different features within a subject is based on the mechanisms of attenuation of the phase being excited by a polychromatic beam of X-ray photons with the help of a detector, which corresponds to the Lambert-Beer Law [23,43].

In preparation for XCT analysis, in order to capture a sufficient

number of GNPs and ensure statistically significant results, cylindrical specimens of hardened mortar with a nominal height of 5 mm and a diameter of 2 mm were extracted from the centre of the $50 \times 50 \times 50 \text{ mm}^3$ cement cubes by core drill for scanning, as seen in Fig. 4(a). All specimens were dried in a vacuum oven at 50°C for 24 h to a constant mass prior to scanning. The cylindrical form and dimensions of the specimens ensured uniformity in radiation penetration and reduced beam hardening artifacts, while achieving the required voxel size.

A high resolution Xradia Versa XRM 520 X-ray CT instrument was used. The voltage and X-ray source of the XCT were set at 80 kV and 125 μA , respectively. The distance between the specimen and the detector was set as 12 mm, as shown in Fig. 4(b). The scanning for each specimen consisted of 3001 projections (rotate from -180° to 180°) and the exposure time was 2 s. Based on the sizes of the scanned specimens, and the dimensions of dry GNPs (20 μm), the output projections were determined in voxel sizes of $1.786 \times 1.786 \times 1.786 \mu\text{m}^3$. The XCT images were analysed using software package Avizo 2020. Before the quantitative analysis, the borders of the reconstructed volume were trimmed to remove any poor-quality images, and a median filter was applied.

16-bit slice images were used as they could display 65536 levels of grey, as opposed to only 256 levels in 8-bit images. This significant increase in greyscale depth allowed for a much finer differentiation of materials, which was critical for accurately identifying GNP particles. Based upon the ranges of grey values of different phases, the different phases could be distinguished, from which the desired geometrical information of GNPs could be extracted. Since the XCT images obtained in this study were in 16-bit, the grey value distribution ranging from 0 to 65535 was exported.

2.4. Mechanical testing of cement mortar

Compressive strength tests were performed at 28 days on a servo-hydraulic test frame. For each mix, three cubic specimens ($50 \text{ mm} \times 50 \text{ mm} \times 50 \text{ mm}$) were prepared in compliance with the ASTM C109 standard [44] to ensure reproducibility.

2.5. Thermogravimetric analysis (TGA)

Thermogravimetric analysis (TGA) using a Mettler-Toledo TGA/DSC1 instrument was conducted to evaluate the decomposition of graphene cement composites and understand the effects of graphene re-agglomeration on the degree of hydration. One sample was analysed for each mix. Prior to testing, samples were ground into powder form and dried at 50°C for 24 h in a vacuum oven. The testing temperature

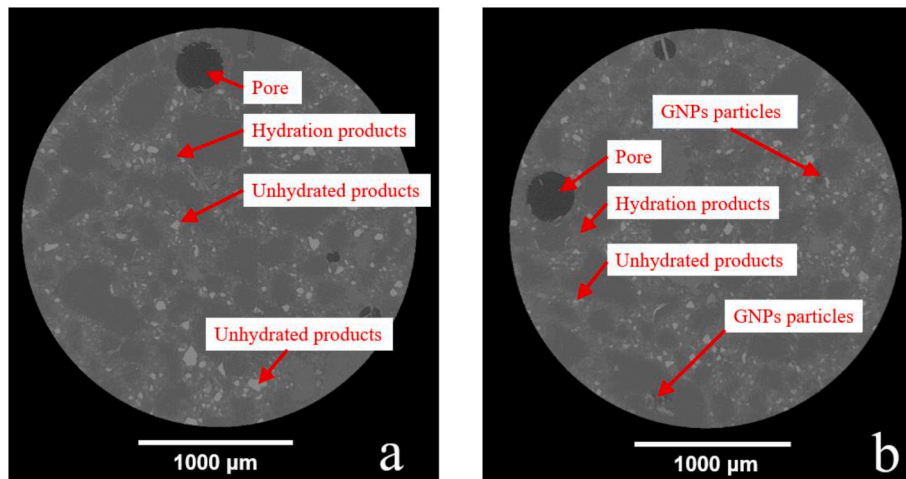


Fig. 5. Different phases with different grey values in (a) reference sample without graphene, and (b) graphene cement composite sample.

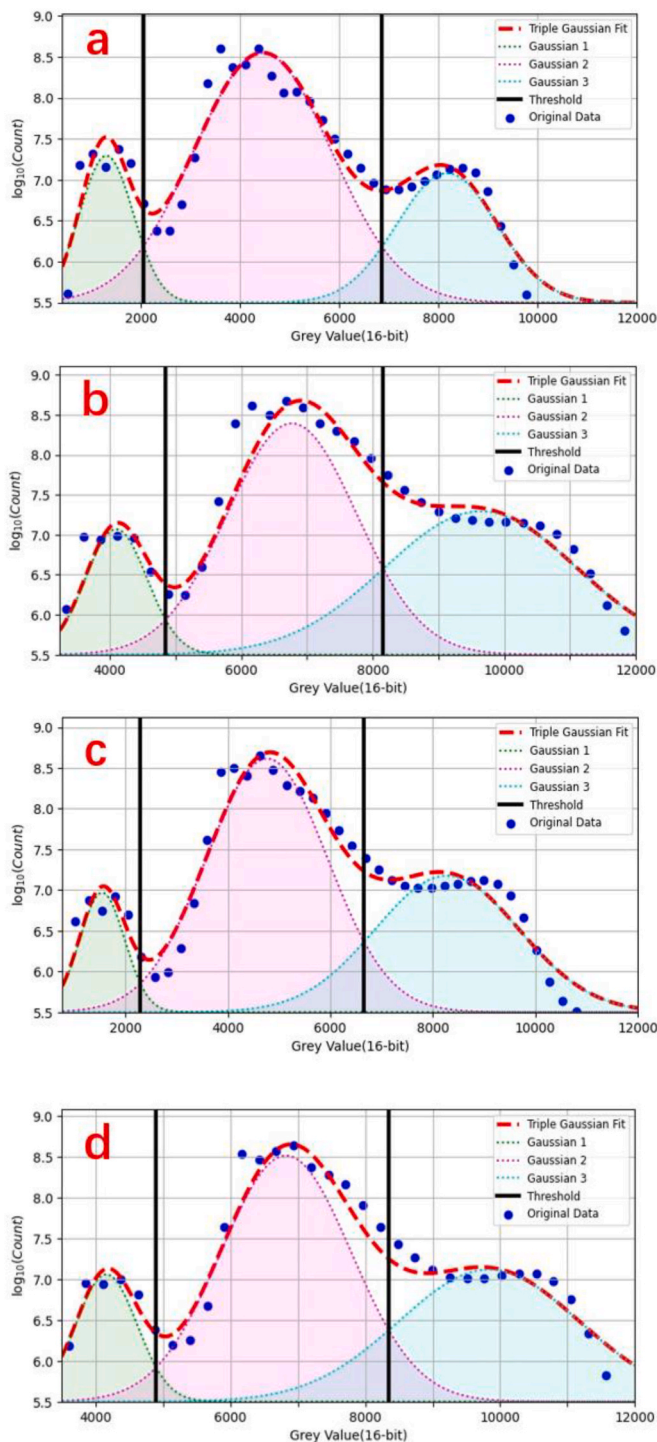


Fig. 6. Grey value distributions of XCT images of all samples for thresholding, showing (a) reference sample G0, (b) G1, (c) G2, and (d) G3.

ranged from 25 °C to 1000 °C with a heating rate of 10 °C/min under nitrogen gas.

3. Results and discussion

3.1. Quantification of grey values for different phases in XCT images

Fig. 5(a) and (b) show typical XCT slice images for mortar samples with and without GNPs. Three phases can be detected in Fig. 5(a): Pores (lowest grey value), hydration products (medium grey) and unhydrated

Table 5

Grey value ranges of pore/GNPs phase for different samples.

Sample	Pore/GNPs phase
G0	≤ 2071
G1	≤ 4700
G2	≤ 2291
G3	≤ 4898

products (brightest area), in addition to sand [23]. Fig. 5(b) contains the additional graphene phase, shown in slender-like form [23,37,45–47].

In order to calculate the volumetric information of the different phases, it is necessary to define the associated greyness. The method presented in Refs. [23,37], where quantitative investigations to determine the grayscale value distributions of XCT slices were carried out to delineate pores, graphene, and solid matrix within graphene modified cement pastes, was adopted in this research.

Fig. 6(a)–(d) show the grey value histograms of the slice images for all samples. Since the focus of XCT image analysis was GNPs and pore phases, these figures presented grey values up to a scale of 12000. The histograms revealed significant peaks, corresponding to the aforementioned phases of the cement composites with and without graphene. Gaussian fitting was successfully applied to the curves. The intersections of the fitted curves were defined as the critical values bounding the different phases. The intersection between Gaussian 1 and Gaussian 2 defined pore/GNPs phases as they featured the similar low densities in the composites [37]. Table 5 summarises the grey values defining the pore/GNPs phases in this study. Grey values in XCT images are affected by sample compositions and measurement conditions, such as temperature and time-dependent instrument drift during prolonged scans. As a result, the exact values always vary between different scans, even for the same sample [37,48]. However, compared to the full range of grey values (0–65535 for 16-bit images), the grey values for the pore/GNPs phase remained within a relatively narrow and consistent range (~2300–4900).

Fig. 7(a) presents an example of the 3D reconstructed volume of G0 (reference sample) after segmentation and labelling, with the colours representing different spatial coordinates only. In this 3D model, large spherical pores can be seen within the cylindrical domain and pores are the only features. On the other hand, the graphene-containing samples (G1, G2, and G3) shown in Fig. 7(b)–(d) exhibited two distinct particle morphologies: Spherical particles and slender platelets. The spherical particles represented pores and the slender, elongated particles were graphene nanoplatelets in line with their two-dimensional, sheet-like structure [2,23,32,35,37,49].

3.2. Determination of threshold for quantitative identification of graphene

Building on the qualitative information generated based on the images shown in Fig. 7, a quantitative method was adopted to identify GNPs from these images. Existing literature on the identification of graphene particles in cement composites using XCT typically differentiates graphene and pore features based on Sphericity [21,37], which is defined as $\frac{\pi^{1/3} \times (6V)^{2/3}}{A}$, where V refers to the 3D volume and A the surface area of the particle according to User's Guide Avizo Software [50]. However, while this parameter indicates deviation from sphericity, it cannot provide a complete description of the particles' slenderness. This simplified morphological criterion may not capture the true complexities and variations in graphene particles, which are characterised by their flat, elongated and two-dimensional nature, leading to potential misclassification and inaccurate quantification of GNPs.

To overcome this limitation, the FeretShape metric was introduced as an alternative, in which the FeretShape value (F-value) is defined as L/l , where l is the minimum Feret diameter and L is the maximum Feret diameter in the orthogonal direction [50]. An F-value close to 1.000

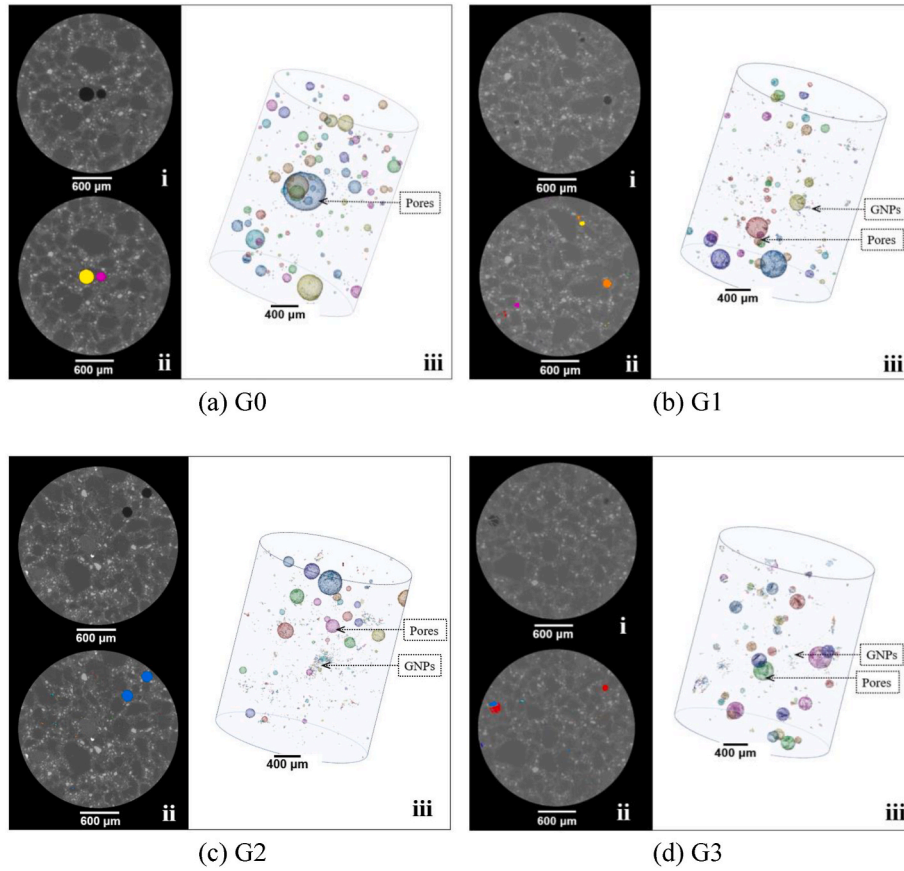


Fig. 7. Identification of 'pore + GNPs' phases in G0, G1, G2 and G3 samples: (i) non-labelled 2D slice; (ii) labelled 2D slice; (iii) reconstructed 'pore + GNPs' structures in 3D volume. In these cases, both pores and slender shape GNPs can be distinguished (for G0 sample, only pores).

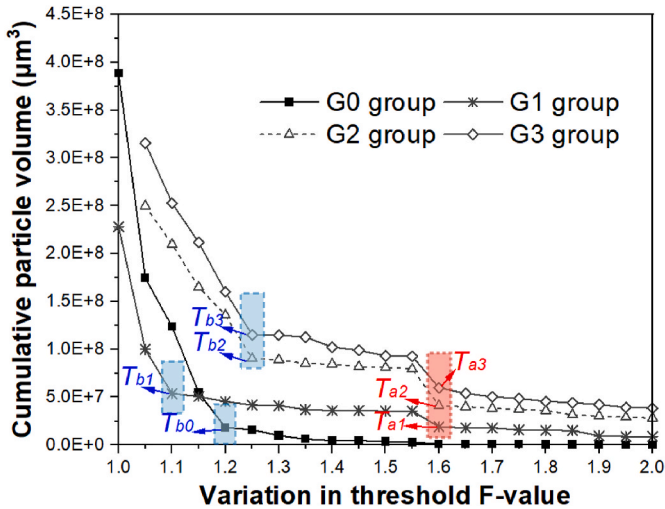


Fig. 8. Cumulative particle volume as a function of variation in threshold F-value.

indicates a spherical object (i.e., pores), while a larger F-value suggests a more slender object (i.e., GNPs). Fig. 8 plots the cumulative particle volume (i.e., pores and GNPs) as a function of different threshold F-values. When the threshold F-value is set at 1.000, it includes all pores and GNPs. As the threshold F-value increases, pores with smaller F-values are excluded. Since GNPs feature a similar slender platelet morphological characteristic, which drastically varies from that of spherical pores [2,23,32,35,37,49], a sudden change in the cumulative

particle volume followed by a plateau is theoretically expected once all pores are excluded.

In Fig. 8, two sets of inflection points were observed, T_a and T_b corresponding to F-values of 1.600 (T_{a1} , T_{a2} and T_{a3}) and 1.100–1.250 (T_{b0} , T_{b1} , T_{b2} and T_{b3}), respectively. It should be noted that the sharp change observed at set T_b was due to the exclusion of a small number of very large spherical pores (as shown in Fig. 9, some on the order of $10^7 \mu\text{m}^3$), rather than a transition from the pores to GNPs phase. On the other hand, at set T_a , the sharp change was attributed to the exclusion of a large amount of small pores and a transition from pore to GNPs phase. This set of points represented the critical points, at which a small increase in the F-value resulted in a sharp reduction in the segmented volume of particles. Thus, T_a can be considered as the F-value threshold separating GNP particles and pore phases. It can be observed that after T_a , there was a prolonged plateau of the cumulative particle volume, which was consistent with the expectation that the total particle volume was that of GNPs after excluding the pores.

To confirm the feasibility of using the F-value at point T_a as the threshold, Fig. 10 shows the total surface area (TSA) of GNPs per unit volume of cement composite (TSA_0), with the deviation bands indicating using different threshold F-values higher than that at T_a . The deviation values in all samples were relatively small.

3.3. 3D imaging of GNPs

Using 2D slice images and the critical F-value, GNPs were isolated for the graphene-containing samples, as revealed by Fig. 11(a)–(c).

The 3D models of graphene particles from the three test samples demonstrated the expected phenomena. G1 contained a low amount of graphene, and Fig. 11(a) indicates a remarkably uniform distribution of GNPs throughout the cementitious matrix, with very limited large-sized

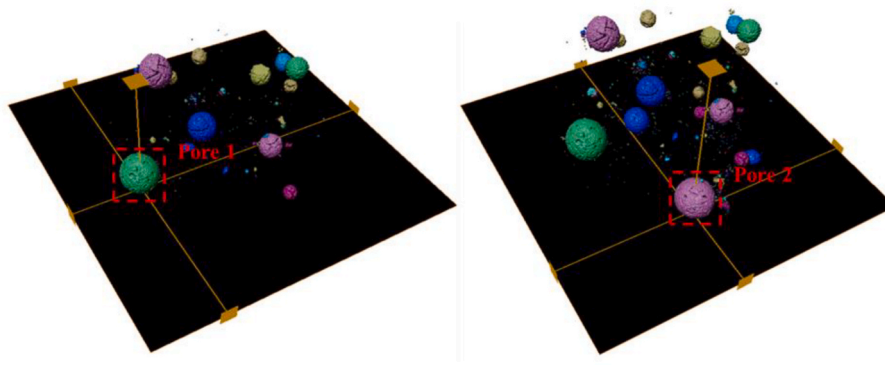


Fig. 9. Example of excluded large spherical pores at F-value greater than set T_b in Fig. 8.

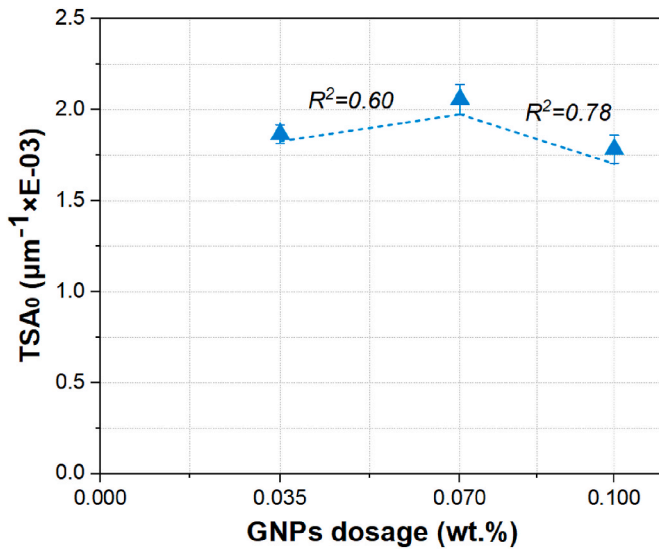


Fig. 10. Variations in TSA₀ vs. dosage of graphene in hardened cement.

GNP agglomerates. In contrast, G3, with a high graphene dosage, demonstrated a high degree of re-agglomeration along with a highly non-uniform distribution of GNPs and notably large GNPs. The images of G2 sample show less re-agglomeration compared to the G3 sample, offering the largest TSA₀ for the formation of hydration products, as shown in Fig. 11(b). Another closer look at a sub-volume for G3 sample in Fig. 12 exhibited re-agglomerated GNP sizes reaching 145 μm in length, while the nominal diameter of the pristine GNPs was 20 μm. The morphology of the reconstructed GNP particles were consistent with those observed by [40,51].

3.4. GNPs size distribution

Fig. 13 presents the results of the size distributions of GNP particles in G1, G2, and G3 specimens. A sieve module was applied to remove GNP particles with a lateral size smaller than the nominal GNPs lateral size of 20 μm. As expected, a higher graphene dosage resulted in a higher degree of re-agglomeration and larger sizes of GNP agglomerates. The maximum GNP agglomerate sizes for G1, G2 and G3 were 122, 151, and 249 μm, respectively. In addition, an increase in GNPs dosage led to a drastic increase in the frequency of large GNP agglomerates, as shown in Fig. 14. This increased number of large-size agglomerates significantly reduced the total surface area of GNPs which can serve as nucleation sites. Their effects on cement properties are discussed in detail in Sections 3.5–3.7.

3.5. Correlation between the total surface area of graphene and compressive strength of cement mortars

Fig. 15 displays the 28-day compressive strengths of cement mortars with varying dosages of graphene. A similar evolution trend was observed as in Table 1: the incorporation of 0.035 and 0.07 wt% GNPs into the cement mortar resulted in compressive strength increases of 12 % and 19 %, respectively, in comparison to plain mortar. However, this upward trend in strength gains reversed upon exceeding a GNPs dosage of 0.07 wt%, indicating this to be the optimal level for enhancing compressive strength.

Fig. 16 compares the TSA₀ values of all GNP particles and percentage increase in the compressive strength of the prepared mixes with respect to the control mix (i.e. G0) at different dosages of GNPs. It is noteworthy that the vast majority of the surface area of GNPs is attributed to their lateral length and width due to their two-dimensional nature. The contribution of the thickness dimension to the total surface area is comparatively small. A direct correlation between the two curves was observed, both peaking at a GNPs content of 0.07 wt%. A statistical analysis showed significant differences between groups ($p = 0.0010$). This clear correlation strongly indicated the key role TSA₀ of GNPs played in determining the optimal graphene dosage from a performance standpoint and can therefore potentially be used as a part of a prediction process on how different mixes will compare against each other in terms of strength. These results were in line with those reported by previous studies [9,17,25], where it was suggested that GNPs offered larger surface area for the nucleation of hydration products, thereby influencing the strength development of cementitious mixes [52–55].

Compared with existing studies (as listed in Table 1), which rely on SEM to qualitatively assess a limited number of large re-agglomerates, this study used XCT to comprehensively quantify TSA₀ and revealed its significant effects on cement strength. Additionally, while previous studies qualitatively attributed the reduced enhancement effect to re-agglomeration, this study is the first to quantify the mechanisms underlying the inconsistent optimal dosages and varying strength improvement rates observed across different studies, linking these to the TSA₀ of graphene as the key determinant in different cement systems. These findings are further discussed in Section 3.6.

3.6. Effect of GNPs re-agglomeration on cement hydration

Fig. 17 shows the TGA and derivative of TG (DTG) results of the prepared mortar mixes after 28 days of hydration. Four major mass losses can be observed, corresponding to the following decomposition patterns: (i) ~30–105 °C for evaporable water and part of bound water; (ii) ~110–250 °C for C-S-H, Ettringite, etc.; (iii) ~350–530 °C for the dehydroxylation of calcium hydroxide (CH); and (iv) 600–800 °C for the decomposition of CaCO₃. The content of CH (%) can be used to evaluate the hydration degree of cement mixes [1,30,52,56].

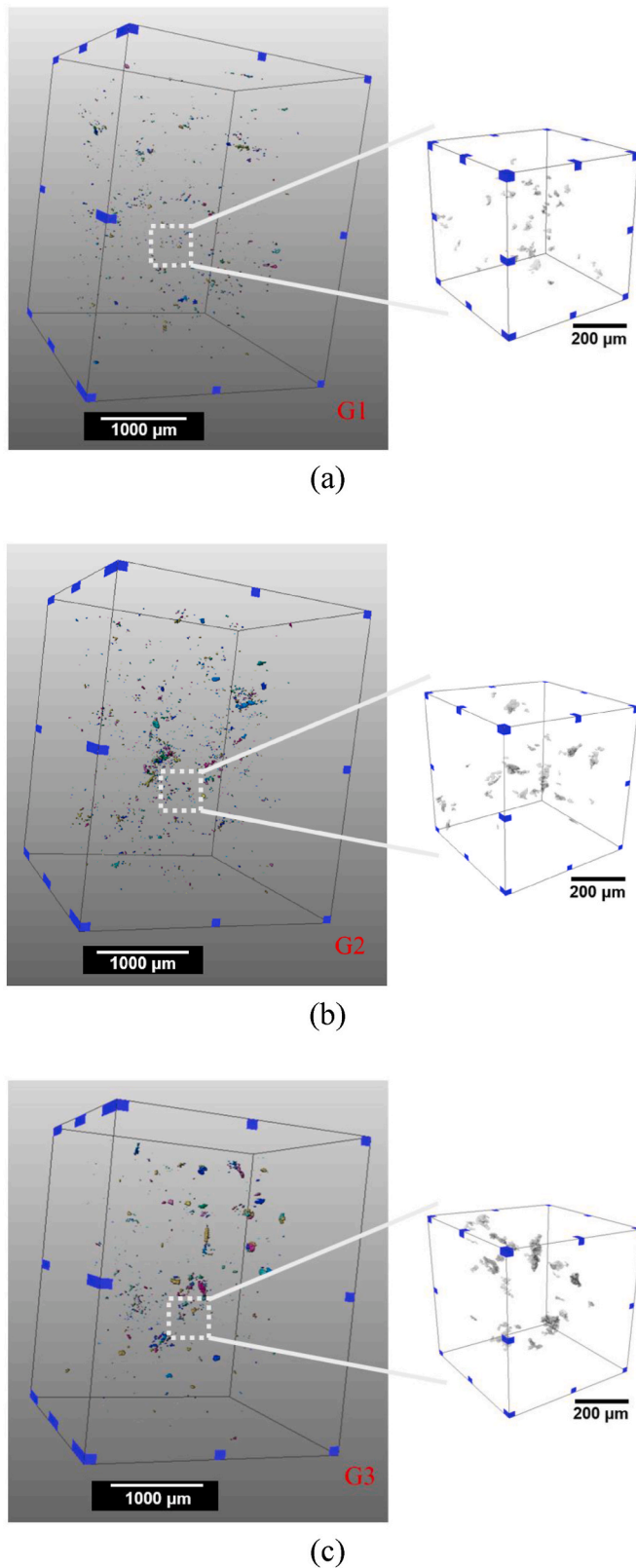


Fig. 11. Distinguished GNP particles' 3D images and close-up view showing samples: (a) G1, (b) G2 and (c) G3.

The results in Fig. 18 show that CH content increased with the addition of GNPs from 0 to 0.07 wt% and then slightly decreased at 0.1 wt% GNPs dosage. The reduction in CH content at 0.1 wt% GNPs dosage was associated with the re-agglomeration of GNPs, as highlighted earlier

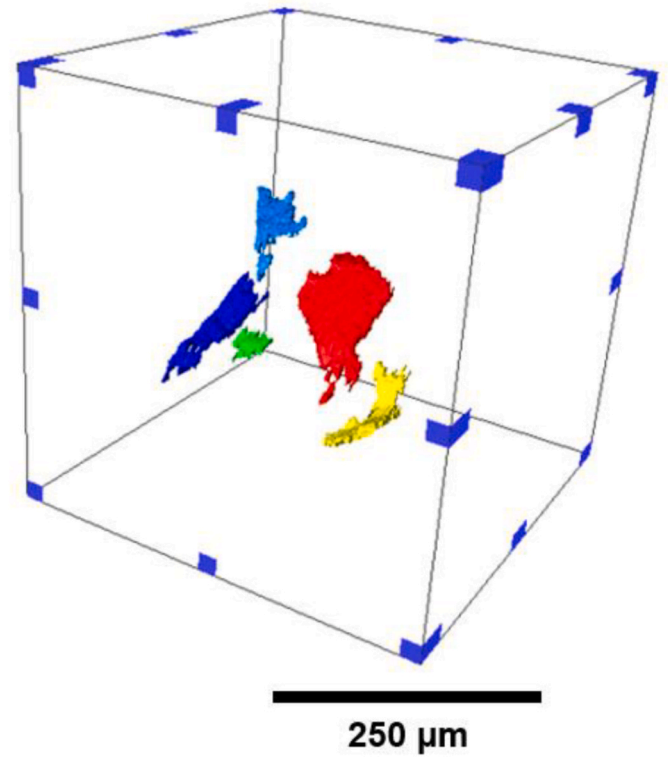


Fig. 12. Large-sized GNP re-agglomerates observed in G3 specimen (extracted sub-volume).

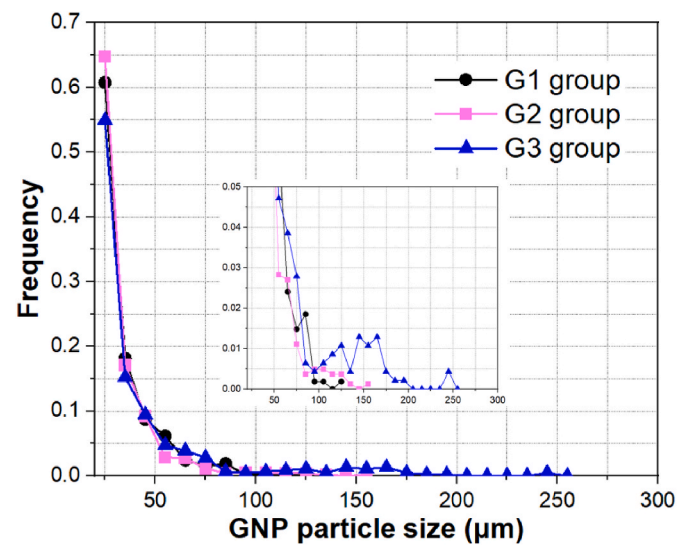


Fig. 13. GNP particle size distribution of G1, G2 and G3 specimens.

and in previous relevant literature [17,29,30]. Since the degree of cement hydration can be inferred from the CH content [45], the higher CH content in graphene-containing samples compared to the control mix indicated the key role GNPs played in promoting the hydration reaction and associated formation of hydrate phases. Moreover, the variation in CH content corresponded well with the change in TSA_0 of GNPs (as depicted in Fig. 16). A statistical analysis revealed a strong positive relationship between TSA_0 and cement hydration ($p = 0.0219$), validating that enhancement in hydration is attributed to the increased surface area provided by GNPs as nucleation sites.

Compared with existing studies that used reduced hydration degree

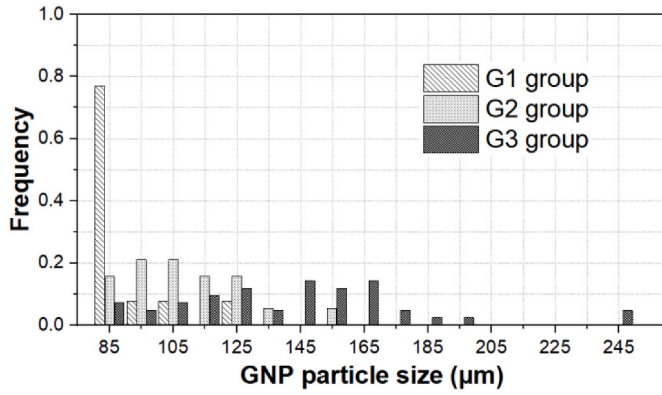


Fig. 14. GNP particle size distribution and frequency in the range of $\geq 80 \mu\text{m}$.

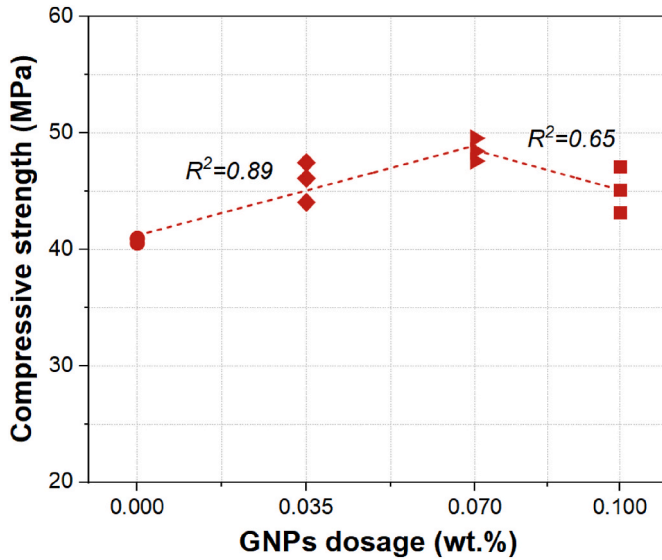


Fig. 15. Correlation between the compressive strength of cement composites and dosage of GNPs.

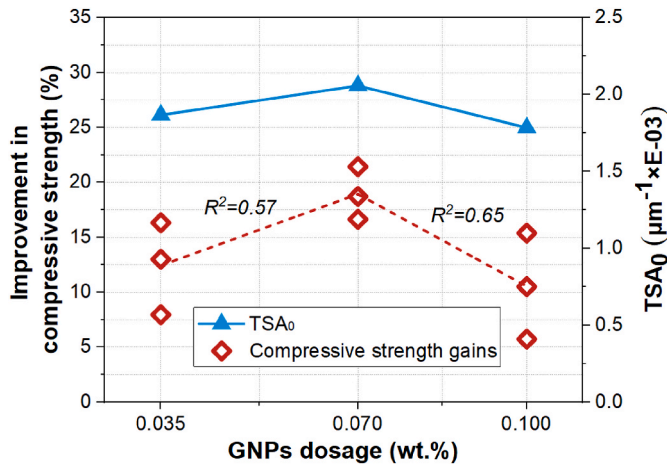


Fig. 16. Correlation between TSA_0 and improvement in compressive strength of cement composites.

or mechanical performance to indirectly assess graphene re-agglomeration in cement-based mixes, this study directly quantified TSA_0 in hardened mixes and correlated it with hydration and strength

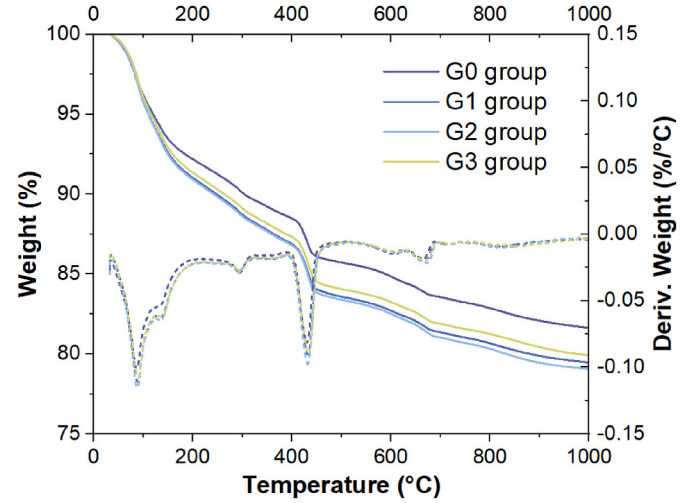


Fig. 17. TGA-DTG curves of graphene modified composites.

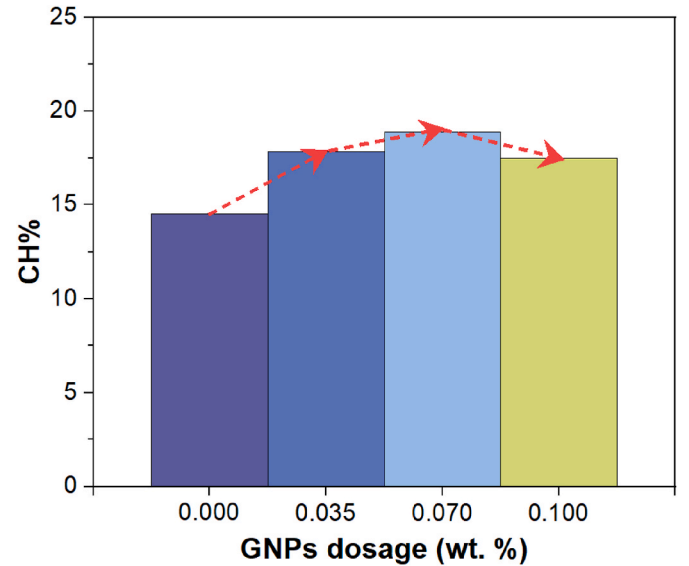


Fig. 18. CH% of cement mixes with different GNPs dosages.

gains. These findings are further explored in Section 3.7.

3.7. Discussions

While there are studies on the potential nucleation effects of graphene in cementitious mixes, this is the first time the quantification of the correlation between the TSA of graphene in hardened mortar and the strength enhancement of cementitious mixes is reported. The results obtained in this study indicate the TSA as the governing parameter for optimal dosage of graphene, enabling cost-effective mixture design and strength prediction.

The hydration process of cement initiates as tricalcium silicate (C_3S) and dicalcium silicate (C_2S) react with water and exchange ions instantaneously [57,58]. In the presence of GNPs, this exchange sees calcium ions (Ca^{2+}) migrating towards GNPs more rapidly than silicate tetrahedra (SiO_4^{4-}), resulting in the rapid adsorption of Ca^{2+} near the GNPs surface [28,59]. Such dynamics are beneficial to the formation of key hydration products such as ettringite (Aft), calcium hydroxide (CH) and calcium silicate hydrate (C-S-H). These hydration products migrate towards GNPs ‘islands’, where Ca^{2+} -rich zones are present. This migration preserves sufficient microchannels for water transport within

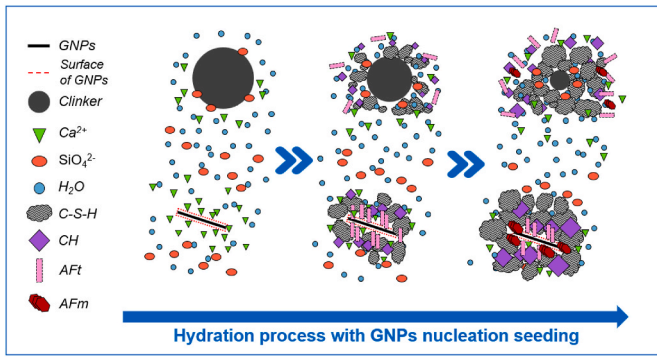


Fig. 19. Illustration of the nucleation seeding effects of GNPs in cement.

cement's hydrolysis layer, promoting ion exchange between the clinker and water [28,60], as demonstrated in Fig. 19. This, in turn, accelerates gypsum consumption, leading to the earlier formation of Aft, and subsequently AFm.

Moreover, GNPs promote the formation of new reaction zones through continuous clinker hydrolysis, enabling more space for C-S-H precipitation, as explained by Refs. [28,60–62]. This phenomenon decreases the diffusion barrier of hydration products surrounding the unhydrated cement particles, resulting in increased levels of hydration. On the other hand, an excessive dosage of GNPs in cement-based mixes can result in their agglomeration, reducing the TSA of the GNPs 'islands' attracting hydration products, with more unhydrated clinker being trapped by hydration products such as Aft and C-S-H, as presented in Fig. 1.

This hypothesis was also validated by the present study and several previous investigations (Fig. 20). Ho et al. [30] reported the changes in the amount of CH determined by thermogravimetric analysis and compressive strength as the dosage of graphene varied. Both the amount of CH and compressive strength peaked at 0.07 wt% GNPs, as shown Fig. 20(b). Cui et al. [29] studied the nucleation seeding effects of GNPs

on the hydration process using QXRD and found that at 56 days, 0.08 wt % GNPs (optimal dosage) led to a 30.8 % increase in the amount of Aft and a 21.7 % enhancement in compressive strength, when compared to the plain cement mix, as shown in Fig. 20(c). In both cases, a direct correlation between the formation of hydrate phases and performance was highlighted, for both of which existed an optimal dosage of graphene that promoted the highest content of hydration products.

Therefore, based on the existing data and the results generated in this study, the optimal reinforcing effect of GNPs can be determined by achieving the largest TSA. This outcome can be further supported by modelling to confirm the optimal dosage of GNPs in various cement-based systems, thereby enabling the efficient design of graphene enhanced concrete mixes involving the optimum graphene content. The findings of this study can contribute to the development of optimal graphene-enhanced concrete mixes for commercial applications, thereby improving mechanical performance via reducing cement usage, and supporting decarbonization in the construction industry.

4. Conclusions

This paper presented the results of a detailed quantitative characterisation study of GNPs in hardened cement mortars via XCT scanning and image processing. The obtained results included spatial location, size and total surface area of graphene in cement mortars containing different graphene dosages. In order to successfully distinguish GNPs from other materials in the XCT images of the hardened cement mortar specimens, grey value analysis was first used to identify pores and GNPs, which have similar grey scales, followed by F-value tests to distinguish between pores and GNPs. TGA and compressive strength tests were also carried out to identify optimum GNPs content and establish a correlation with the total surface area of GNP particles within the prepared mixes.

The main conclusions of this investigation are as follows:

- An F-value threshold of 1.600 can be used to distinguish between pores and GNPs in cement mortar when processing XCT images.

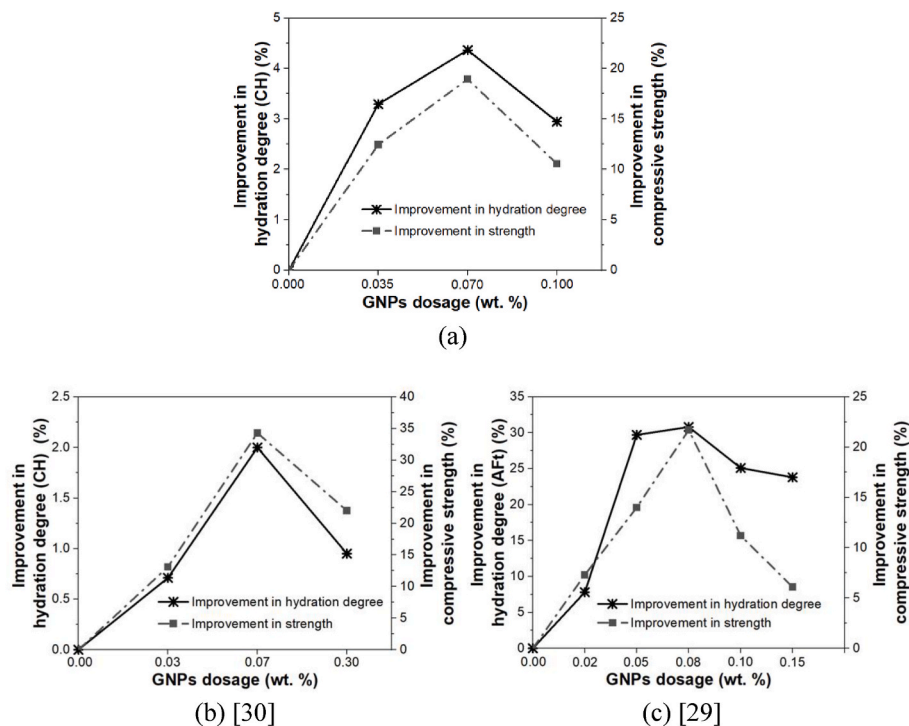


Fig. 20. Correlation between GNPs dosage, hydration degree and enhancement in compressive strength of cement mix, based on the results in: (a) this study; (b) and (c) relevant literature.

- Amongst the three graphene dosages of 0.035, 0.07 and 0.1 wt%, the observed maximum lateral sizes were 122, 151, and 249 μm , respectively. The observed maximum lateral size of GNP agglomerates can reach over 200 μm , suggesting severe re-agglomeration of graphene at high graphene dosages.
- Within the mixes prepared under this study, the optimal graphene dosage is found to be 0.07 wt%. This optimal value was associated with the largest TSA₀ of GNPs in the sample, corresponding to the highest amount of nucleation seeds for cement hydration.
- There was a clear correlation between the total surface area of graphene per unit volume of cement mortar and the increase in hydration degree and compressive strength of cement mortar. This can be an indication of the role graphene plays in the enhancement of hydration reaction.
- Based on the correlation demonstrated in this study, the total surface area of graphene in cement mortar can be used to determine the optimal dosage of graphene in cement-based mixes.

Overall, this study has, for the first time, presented the use of an XCT-based characterisation methodology to quantify the distribution and re-agglomeration of GNPs and correlate the total surface area with the enhancement effect on mechanical performance. This novel approach has provided quantitative evidence to underlie the hypothesis of the nucleation seeding effects of graphene in cementitious materials, which has so far only been qualitatively observed by previous studies. Results obtained here have clearly demonstrated the governing effects of the total surface area of graphene nanoplatelets in enabling the optimum design of graphene enhanced concrete mixes, which significantly differs from the currently adopted mostly trial-and-error based lab activities. The results from this study can serve as a baseline for future optimisation of graphene-cement composites for high-performance applications in civil engineering, and will lay the foundation for numerical models to predict the optimal dosage of GNPs within the cementitious matrix in terms of material cost and performance, further advancing the commercialisation of novel concrete formulations for a range of applications. Furthermore, these findings contribute to achieving sustainability by promoting resource-efficient mix designs and reducing the carbon footprint of concrete production.

CRediT authorship contribution statement

Zhaohua Wang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Meini Su:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Yong Wang:** Writing – review & editing, Supervision. **Cise Unluur:** Writing – review & editing. **Suning Li:** Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Meini Su reports financial support was provided by Engineering and Physical Sciences Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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